

Synthetic showcase matrix

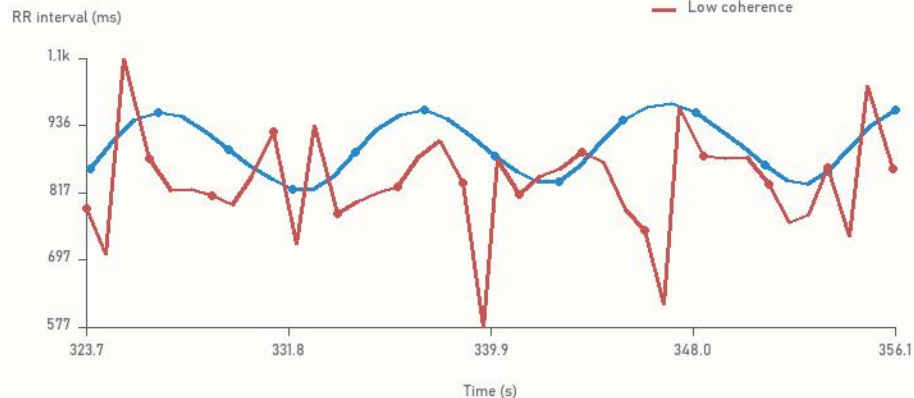
Deterministic showcase-v1 scenarios with publication-facing summary telemetry exported from SyntheticBio.

Scenario overview					
Scenario	Coherence (0-1)	RMSSD (ms)	Interval SampEn	Amplitude SampEn	Primary state
Breathing pause	0.310	9.6	0.569	0.349	Stale
Coherence high	0.517	27.5	0.000	--	Tracking
Coherence low	0.286	130.8	0.991	1.099	Tracking
Entropy high	0.300	13.8	1.230	0.426	Stale
Entropy low	0.327	8.5	0.044	0.148	Stale
Entropy rising	0.324	13.3	0.890	0.445	Stale
Flat breathing	0.268	3.3	--	--	Waiting for breathing tracking
HRV high	0.107	58.5	0.514	0.938	Tracking
HRV low	0.335	3.0	0.015	0.189	Tracking
Irregular RR	0.200	186.0	1.429	0.301	Tracking
Entropy jittered	0.257	13.3	1.174	0.426	Stale
10 bpm	0.502	22.8	0.000	1.303	Tracking
12 bpm	0.445	17.3	0.625	0.393	Tracking
18 bpm	0.181	11.6	0.537	1.217	Tracking
24 bpm	0.265	6.6	0.379	0.823	Tracking
Regular	0.322	9.7	0.190	0.422	Tracking
Resonance (0.10 Hz)	0.516	27.5	0.000	3.135	Tracking

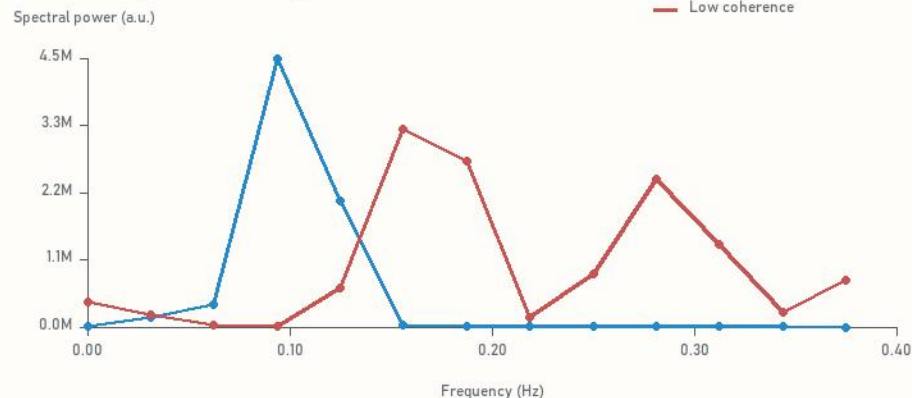
Coherence derivation

Accepted RR intervals are resampled into a tachogram, transformed to a power spectrum, and summarized with both the cited paper ratio and the app-facing 0-1 scores.

A. Accepted RR interval window



B. Power spectral density



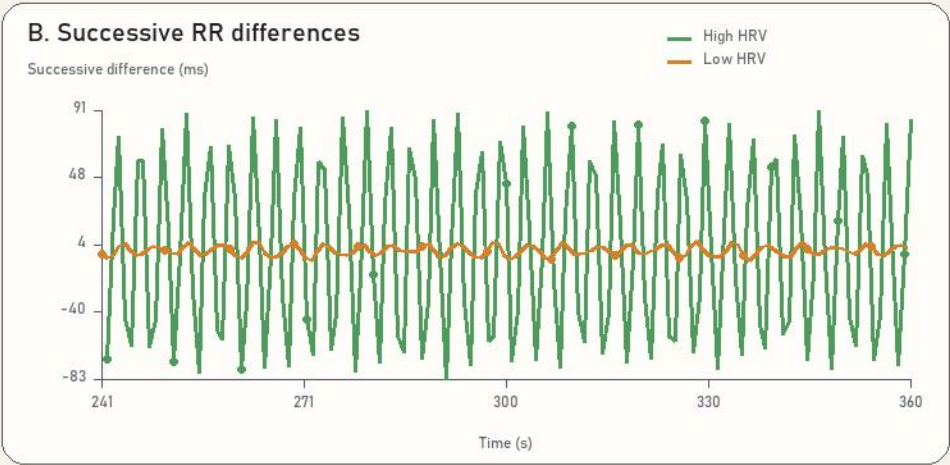
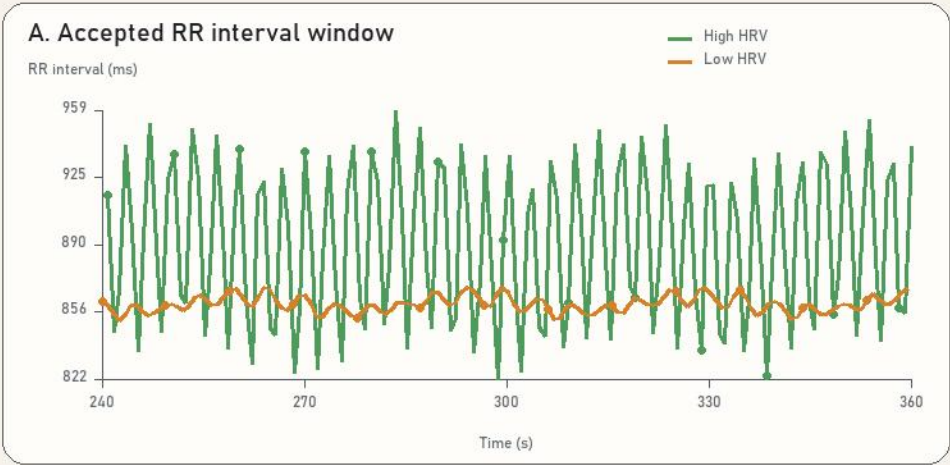
C. Coherence summary

Scenario	Peak freq. (Hz)	Peak window (a.u.)	Total band (a.u.)	Paper ratio	Normalized (0-1)	Displayed (0-1)
High coherence	0.097	124217	224684	1.529	0.553	0.517
Low coherence	0.167	102350	434405	0.095	0.236	0.286

Paper ratio = (peak-window power / (total band power - peak-window power))^2. Normalized coherence = peak-window power / total band power. Displayed coherence is the app-facing 0-1 readout.

HRV derivation

Accepted RR intervals are differenced beat-to-beat and summarized with short-term time-domain HRV metrics following the Shaffer-Ginsberg framing.



C. Time-domain HRV summary

Scenario	RMSSD (ms)	SDNN (ms)	pNN50 (%)	ln(RMSSD)
High HRV	58.5	40.0	59.0	4.069
Low HRV	3.0	4.1	0.0	1.108

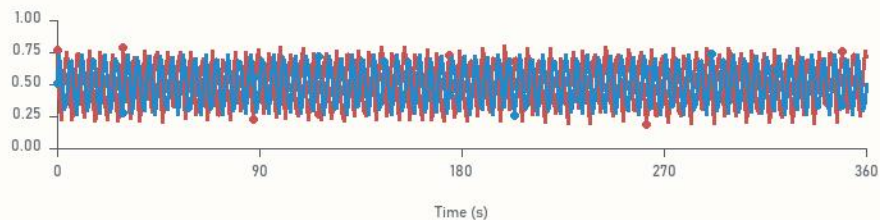
These traces are intended as reproducible short-term examples. They are not substitutes for 24 h norms or clinical artifact adjudication.

Breathing-dynamics derivation

An ACC-derived breathing waveform is converted to accepted extrema, breath-interval and breath-amplitude series, and entropy-oriented summary statistics.

A. Breathing waveform

Normalized amplitude (a.u.)



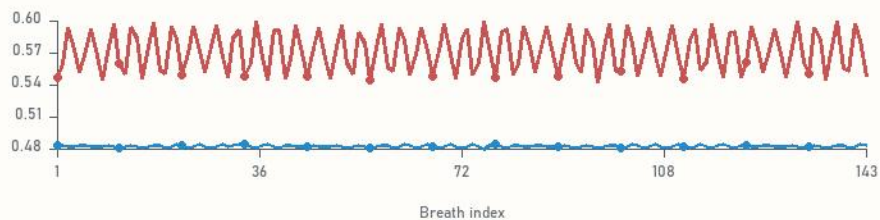
B. Breath-interval series

Interval (s)



C. Breath-amplitude series

Amplitude (a.u.)



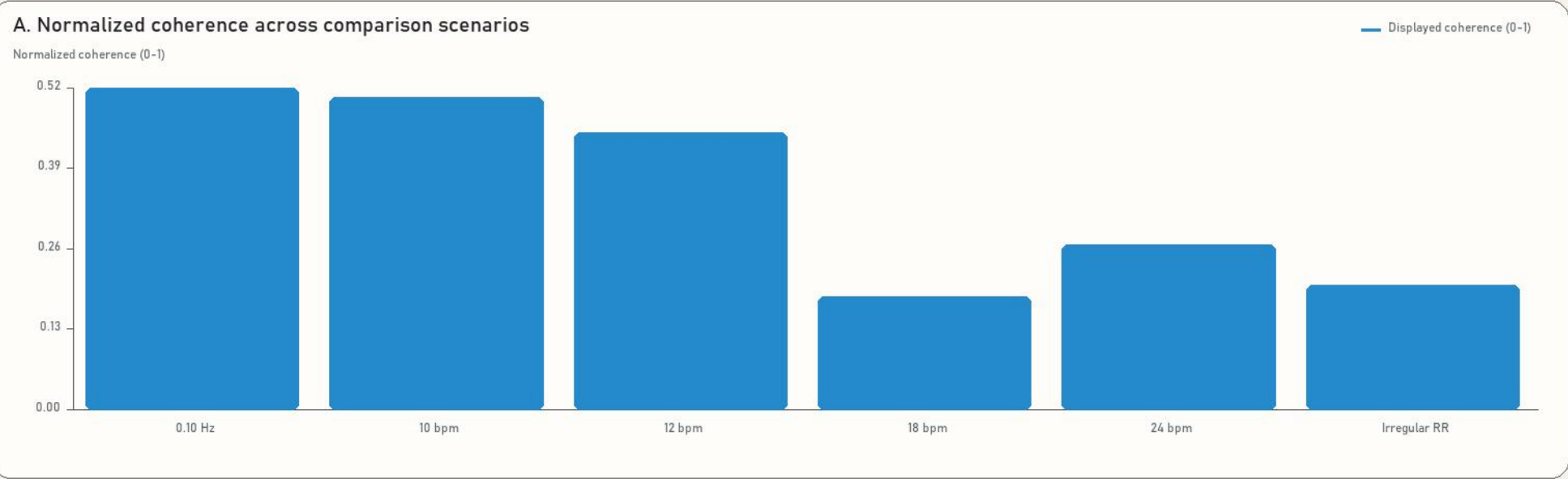
D. Entropy and variability summary

Scenario	Interval SampEn	Amplitude SampEn	Interval CV	Amplitude CV
High	1.230	0.426	0.020	0.034
Low	0.044	0.148	0.008	0.002

SampEn settings: $m = 2$, $\text{delay} = 1$, $r = 0.2 * \text{SD}$. The waveform source is the repo's calibrated ACC breathing signal rather than a respiration belt.

Coherence appendix

The resonance condition remains separated from the off-resonance sweep and the irregular-RR comparison scenario.



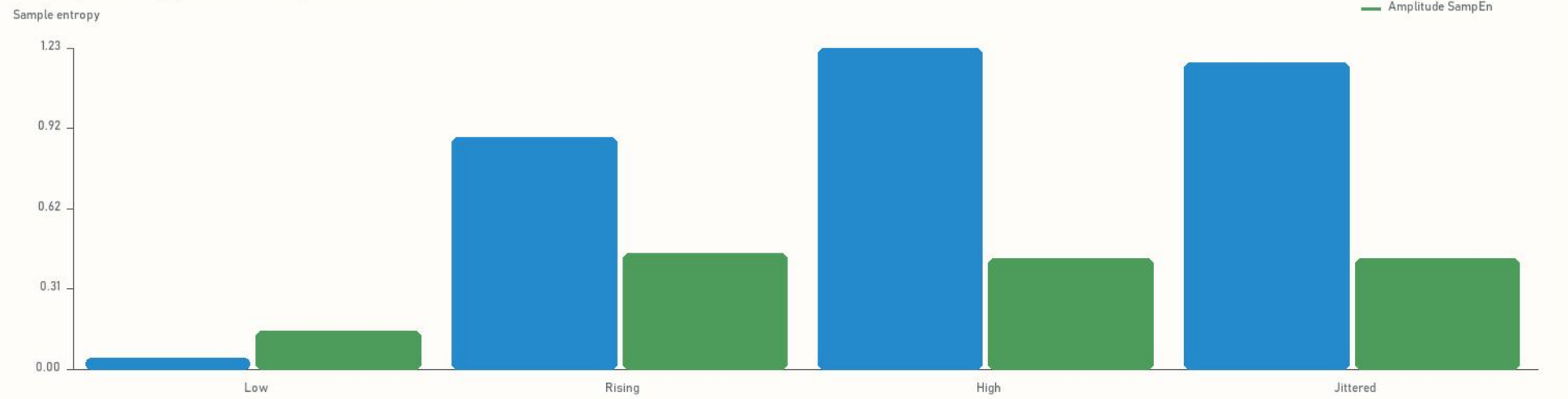
B. Scenario summary

Scenario	Peak freq. (Hz)	Paper ratio	Displayed (0-1)
0.10 Hz	0.097	1.519	0.516
10 bpm	0.163	0.825	0.502
12 bpm	0.198	0.664	0.445
18 bpm	0.040	0.049	0.181
24 bpm	0.040	0.193	0.265
Irregular RR	0.164	0.099	0.200

Dynamics appendix

Entropy variants and tracker edge states are retained in the supplementary bundle for comparison and failure-mode reporting.

A. Sample entropy across comparison scenarios



B. Edge-state summary

Scenario	Tracking state	Interval SampEn	Amplitude SampEn
Flat breathing	Waiting for breathing tracking	--	--
Breathing pause	Stale	0.569	0.349